

4. Include statistical analysis components:

- Calculate and display key summary statistics (mean, median, standard deviation, skewness, kurtosis)
- Test for stationarity in both price and return series
- Calculate and visualize autocorrelation in returns
- Identify and highlight outlier days and their potential causes

5. Add interactive elements (if using a tool like Jupyter):

- Allow selection of different time periods for analysis
- Include options to normalize data in different ways (percent change, z-score, etc.)
- Provide capabilities to zoom into specific time periods of interest

This project will develop your skills in comprehensive financial data visualization and statistical analysis, helping you identify patterns and relationships that could form the basis of trading strategies.

Project 12: Predictive Time Series Analysis System

Objective: Build a system that applies time series decomposition and forecasting techniques to financial data, evaluates their effectiveness, and identifies potential trading signals.

Requirements: